

Home Paper Submission-IV

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Design of a plant to produce 20,000 TPA of 2-hydroxyethyl methacrylate

by

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Chapter 1: Introduction

2-Hydroxyethyl Methacrylate, abbreviated as **HEMA**, is a hydroxy ester of methacrylic acid. It is a monomer that is used to make hydrogels used in making UV cured coatings and adhesives. The polymer (poly Hydroxyethyl Methacrylate or **PHEMA**) is used to make hydrogels and contact lenses and are used for drug delivery in biomedical fields. Since the monomer polymerizes quite easily, hydroquinone or methoxyphenol are added as inhibitors.

1.1 Product definition and properties

Name: 2-Hydroxyethyl Methacrylate (HEMA)

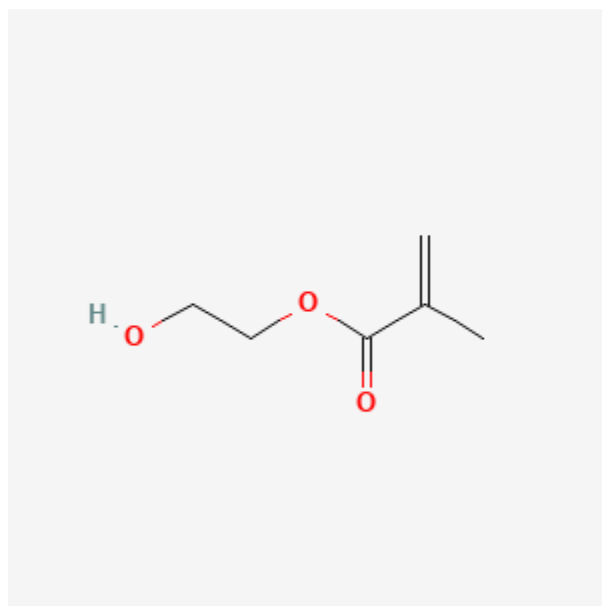


Figure 1: Structure of 2-Hydroxyethyl Methacrylate

Molecular formula: C₆H₁₀O₃

Appearance: Colourless, viscous liquid

CAS No: 868-77-9

IUPAC name: 2,2-hydroxy,ethyl-methacrylate

Other names: 2-hydroxyethyl 2-methylprop-2-enoate, Glycol methacrylate, and ethylene glycol methacrylate, Glycol mono-methacrylate

Table 1: Physico-chemical Properties of HEMA

Property	Value
Molecular Weight	130.14 g/mol
Density (@ 20°C)	1.07 g/cm ³
Melting point	-12 °C
Boiling point (@1 atm)	205 °C
Flash point (@1 atm)	107 °C
Solubility in water	Completely miscible
Vapor pressure (@ 20°C)	20 Pa

1.2

Hazards and Toxicity (Sigma-Aldrich, 2006)

Signal word: Warning.

Hazard statements: Causes skin irritation. May cause an allergic skin reaction. Causes serious eye irritation.

Precautionary statements: Prevention: P261 Avoid breathing mist or vapors. P264 Wash skin thoroughly after handling. P280 Wear protective gloves/ eye protection/ face protection. Response: P333 + P313 If skin irritation or rash occurs: Get medical advice/ attention. P337 + P313 If eye irritation persists: Get medical advice/ attention. P362 + P364 Take off contaminated clothing and wash it before re-using.

1.3 Uses of HEMA

The industrial and biomedical applications of HEMA (2-hydroxyethyl methacrylate) encompass a broad spectrum due to its versatile chemistry and properties. HEMA's high hydrophilicity and polymerizability make it indispensable in developing materials where mechanical strength, biocompatibility, and aqueous interaction are critical.

1.3.1 Industrial Applications

1. Synthesis of coating resins for automotive, architectural, and paper industries
2. Production of advanced adhesives and sealants
3. Textile finishing agents to improve fabric durability and water resistance
4. Printing inks for high-performance applications
5. Manufacture of hygiene products and plastic modifiers
6. Components in UV-curable and photopolymerizable formulations

1.3.2 Biomedical Applications

1. Contact lens hydrogels due to excellent water absorption and biocompatibility
2. Biosensors and membranes leveraging its inertness and hydrogel formation
3. Dental restorative materials such as adhesives and composite resins for hydrolytic and enzymatic stability
4. Controlled drug delivery systems using hydrogel networks for sustained release
5. Artificial organs and tissue engineering scaffolds
6. Cell culture substrates facilitating spheroid formation in biotechnology research

1.4 Market Analysis

Global production of 2-Hydroxyethyl Methacrylate (HEMA) is estimated to be around 300,000–350,000 tonnes per annum. The global market size in 2024 is valued at approximately USD 1.2 billion, with a compound annual growth rate (CAGR) of 6.5% for the forecast period 2024–2030. HEMA demand is primarily driven by its use in adhesives, coatings, resins, UV-curable materials, and contact-lens formulations.

According to trade statistics and industry reports, India currently has no domestic manufacturers of HEMA, and the country is entirely import-dependent. Imports during 2023–2024 are estimated at 7,000–8,000 tonnes per annum, mainly sourced from China, Japan, and Europe. Based on average import prices ranging between USD 2.5–10.5 per kg, the annual import value is estimated between USD 18 million and USD 84 million. There is negligible export activity from India, as all imported volumes are consumed domestically for downstream resin and coating applications. The exports are a very negligible quantity.

Globally, the largest manufacturer is Wanhua Chemical Group, operating a plant with a capacity of 39,500 tonnes per annum in China. Other significant producers include Dow Chemical Company (USA) and Mitsubishi Chemical Corporation (Japan). Smaller contributions come from Evonik Industries, Nippon Shokubai, and GEO Specialty Chemicals.

Due to the absence of indigenous manufacturing, the Indian HEMA market is driven by imports, meeting demand from local resin, paint, and polymer industries. Given the current market size and a healthy CAGR, India represents a potential opportunity for setting up a 10,000–40,000 TPA domestic production facility. Such a plant could substitute imports, reduce foreign exchange outflow, and meet the growing needs of coatings and specialty polymers sectors.

The price of HEMA varies significantly based on purity and grade. Technical-grade HEMA (used in coatings and adhesives) trades at around USD 2.5–4.0 per kg, whereas high-purity grades (for biomedical and optical applications) can exceed USD 10 per kg. Prices are sensitive to fluctuations in feedstock costs of methacrylic acid and ethylene oxide, as well as logistics and polymerization inhibitor requirements.

In conclusion, the Indian HEMA market is poised for steady growth in line with global trends. With increasing demand for adhesives, coatings, and advanced polymer materials, establishing domestic HEMA production presents a viable opportunity. This could position India as a regional supplier in South and Southeast Asia in the long term.

Table 2: Imports of HEMA

Quantity (MT)	Total value (million USD)	Value per kg	Country
444.54	70.75	160.00	Singapore
398.78	61.4	153.00	China
119.63	19.54	163.0	Japan
4.8	2.26	471.0	Switzerland
0.2	0.15	745.0	South Korea

1.5 Capacity of the HEMA manufacturing

The proposed capacity for the new 2-Hydroxyethyl Methacrylate (HEMA) plant is set at **20,000 tonnes per annum (TPA)**. At present, India imports approximately 7,000–8,000 tonnes of HEMA every year, as there are no indigenous producers. Since the estimated CAGR for HEMA is 6.5%, setting up a plant with capacity of 20,000 TPA, which is approximately half of the largest producer's capacity, the plant can be economically feasible and it will be easy to sustain the market.

Chapter 2: Process selection

2.1 Literature survey

Upon doing a literature survey, it was found that two major industrial processes are used to produce HEMA economically. The first mention of the synthesis is dated to 1925. The two common methods of synthesis are (i) Reaction of Methacrylic acid with Ethylene Glycol and (ii) Reaction of Methacrylic acid with Ethylene Oxide (). Both reactions can lead to formation of Ethylene Glycol Di-methacrylate (abbreviated as **EDGMA**). Other reactions such as acyl chloride esterification method and dihydroxyl carbonic esterification reactions are also possible reactions to produce HEMA, however the conditions required for these reactions are achievable only at lab scale.

2.1.1 Epoxidation of Methacrylic Acid

Upon reacting ethylene oxide with methacrylic acid, the reaction producing HEMA proceeds by a nucleophilic substitution reaction at the (alpha) carbon. The reactivity of ethylene oxide plays a significant role in designing the process since it is a highly reactive and toxic compound. Hence the reaction is conducted at controlled temperatures and pressures. The common choices for a solvent media are any tertiary amines.

2.1.2. Trans-esterification of Ethylene Glycol

An alternative route involves the esterification of methacrylic acid with ethylene glycol. In this process, methacrylic acid reacts with an excess of ethylene glycol in the presence of an acid catalyst—commonly sulfuric acid, p-toluene sulfonic acid, or an acidic ion-exchange resin. The reaction proceeds via condensation, forming 2-hydroxyethyl methacrylate and water as a by-product. The formation of water is the major drawback of this method, since HEMA is completely miscible in water. Water also acts as a polymerization promotor and hence it leads to the formation of hydrogel.

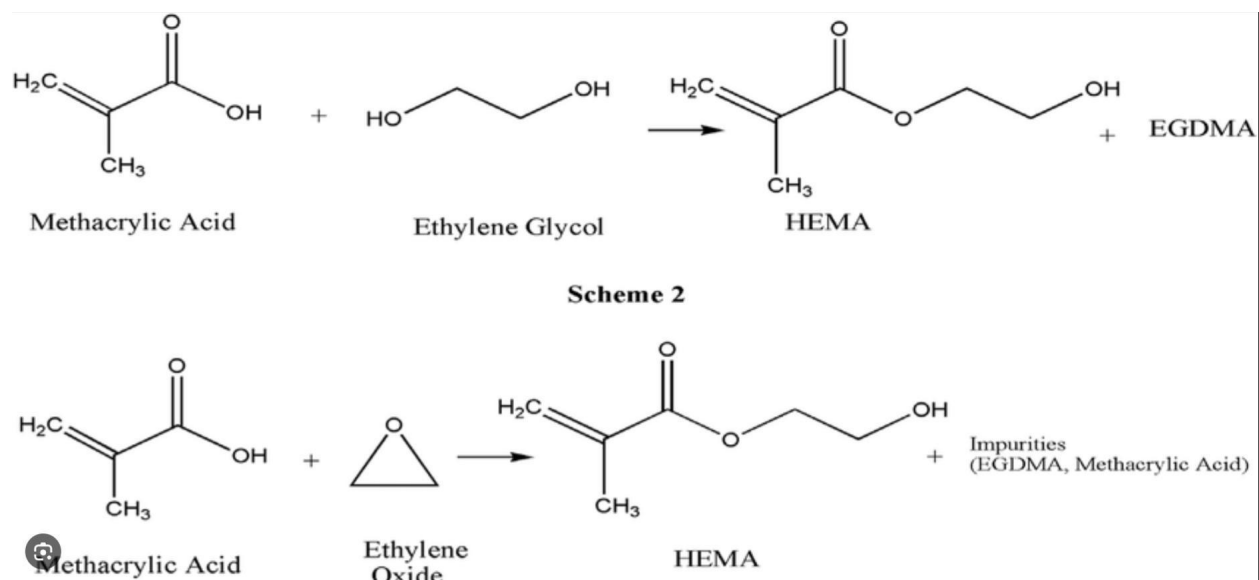


Figure 2: Synthesis routes for HEMA

2.2 Comparison of the two processes

Table 3: Comparison of synthesis routes for HEMA

Process Variable	Esterification Reaction	Epoxidation Reaction
Pressure	Atmospheric Pressures	Vacuum of -0.99 – -0.75 Bar
Temperature	100-120 °C	75 -90 °C
Catalyst	P-methyl benzene sulfonic acid or ion exchange catalysts	Ferric Formate or Chromium Acetate
Reagent	Sulphuric acid	Tertiary amine such as pyridine, N,N-Dimethyl toluene etc.
Reaction time	1 – 3 hr	1.5 – 2 hr

Percentage yield	> 85%	> 99%
Polymerization inhibitor	Phenothiazine	MEHQ (hydroquinone monomethyl ether)
Impurities	Ethylene glycol dimethacrylate, Water	Ethylene glycol dimethacrylate

2.2.1 Analysis of the Esterification reaction (Changzhou Tianma group, 2012)

Advantages of the reaction are as follows-

1. The reaction is relatively simpler and it does not require using complex and hazardous raw materials, which also makes it safer than the ring opening reaction.
2. The reaction conditions are moderate.
3. There is no need to use expensive metal oxide catalysts.
4. Since ion-exchange resins can be used, the separation can be done much more easily.

Disadvantages of the reaction are as follows-

1. The reaction is limited by equilibrium.
2. The amount of side product (EGDMA) produced is much higher even if the amount of acid is controlled. There is a high change of a co-polymer formation if the amount of EGDMA is not controlled.
3. The water formed in this reaction is a major issue since HEMA is highly soluble in water. Water also promotes the polymerization reaction.
4. The yield and selectivity is also lower for this reaction

2.2.2 Analysis of Epoxidation of Methacrylic acid (Changzhou Hickory Chemistry Ltd. 2016)

Advantages of the reaction are as follows-

1. The reaction has better yield and selectivity and the amount of side-product is less.
2. There is no water formed as it is a direct one step synthesis.
3. The purification hence becomes simpler.

Disadvantages of the reaction are as follows-

1. Ethylene oxide is a highly flammable, highly carcinogenic and reactive material.
2. The reaction requires a very controlled environment. If the pressures or temperatures deviate even a little, there is a very high chance of a runaway reaction and a very major accident.

2.2.3 Comparison of the Routes based on economic potential

Economic potential is essentially the net difference between the price of the raw materials and the product. It can be used to compare the economic feasibility of two reactions to synthesize a chemical product.

The economic potential (EP) of a chemical product is given by the following formula

EP = Value of Products – Cost of Raw Materials and Utilities

or equivalently,

$$EP = \Sigma(m_p \times P_p) - [\Sigma(m_r \times P_r)]$$

Where-

m_p is the Mass (or molar) flow rate of product p

P_p is the Price of product p (Rs/kg or \$/kg)

m_r is the Mass (or molar) flow rate of raw material r

P_r is the Price of raw material r

Using this, we can calculate the economic potential of the two routes. The route with more EP will be more feasible.

Table 4: Economic Potential of HEMA based on route using Ethylene Glycol

Chemical	Moles	MW	Price/kg	price/mole
EG	1	62	65	4.03
MAA	1	86	150	12.9
H2O	1	18	0	0
HEMA	1	130	160	20.8

For the esterification reaction,

$$EP = 20.8 - 12.9 - 4.03$$

$$EP = 3.87$$

For the esterification reaction,

$$EP = 20.8-12.9-4.03$$

$$EP = 3.87$$

Table 5: Economic Potential of HEMA based on route using Ethylene Oxide

Chemical	Moles	MW	Price/kg	price/mole
EO	1	44	86	3.784
MAA	1	86	150	12.9
HEMA	1	130	160	20.8

For the epoxidation reaction,

$$EP = 20.8-12.9-3.78$$

$$EP = 4.12$$

2.2.4 Final selection of the process

Based on the factors such as economic potential, raw materials used, yield, selectivity and the reaction conditions, the epoxidation route is more feasible for bulk production of HEMA. This route is also preferred globally.

2.3 Production of Methacrylic acid

Setting up a methacrylic acid (MAA) manufacturing plant alongside a 2-hydroxyethyl methacrylate (HEMA) plant is necessary for both economic and technical reasons. Economically, methacrylic acid is a key raw material for HEMA production through epoxidation; producing MAA onsite reduces raw material procurement costs, transportation expenses, and supply chain risks associated with market fluctuations or delays. This integration improves overall cost efficiency by enabling economies of scale in raw material handling, utilities, and infrastructure usage. Technically, the close coupling of methacrylic acid production with HEMA manufacturing streamlines process flow, enhances material purity control, and minimizes losses during intermediate transfer. Additionally, onsite MAA production provides flexibility in adjusting HEMA feed composition and scaling production volumes rapidly based on market demand. Together, these factors support a more robust, cost-effective, and technically optimized specialty monomer manufacturing facility essential for maintaining competitiveness in rapidly growing biomedical, coatings, and industrial segments where HEMA is widely employed.

A literature survey shows three major industrially relevant routes for producing methacrylic acid (MAA):

1. C4 oxidation (oxidation of C4 hydrocarbons such as isobutene/isobutane to methacrolein/methacrylic acid)
2. ACH (Acetone Cyanohydrin) process (conversion of acetone via acetone cyanohydrin to methacrylamide and then to MAA/MMA)
3. Oxidative dehydrogenation/dehydration of isobutyric acid (IBA) to MAA.

All three routes have been studied and commercialised to varying extents. The ACH route has been the historical commercial route for MMA/MAA, while C4 oxidation routes (via isobutene → methacrolein → MAA) and IBA oxidative dehydrogenation/dehydration have been investigated and implemented depending on feedstock availability and economics.

2.3.1 C4 Oxidation Route

The C4 oxidation route typically oxidises C4 streams (isobutene, isobutane, or mixtures from steam

crackers) to methacrolein (MAL) and subsequently to methacrylic acid (MAA). It is a controlled catalytic oxidation of the C4 rich raw material. (Roma Chemical Ltd, 2006)

Typical catalysts & conditions

1. Catalysts: vanadium-phosphorus based catalysts (e.g., VPO, vanadium pyrophosphate), mixed metal oxides (Mo–V–P, V–Mo–Nb), and supported vanadium oxides.
2. Reaction steps: partial oxidation of isobutene/isobutane to methacrolein (gas-phase, oxidative catalysts), followed by oxidation of methacrolein to MAA.
3. Temperatures: commonly in the range ~300–450 °C for gas-phase oxidation catalysts for direct IBA oxidation; partial oxidation steps for C4 → methacrolein operate at lower temperatures (200–350 °C) depending on catalyst.
4. Pressure: near atmospheric to a few bar depending on reactor design.

Yields & selectivity

Industrial C4 oxidation processes can achieve high conversions with varying selectivity to MAA/MAL depending on catalyst and process optimisation. Reported selectivities to desired C4 oxidation products (MAL/MAA) are high but side oxidation to CO_x and other by-products occurs and must be minimized with appropriate catalyst and reactor design.

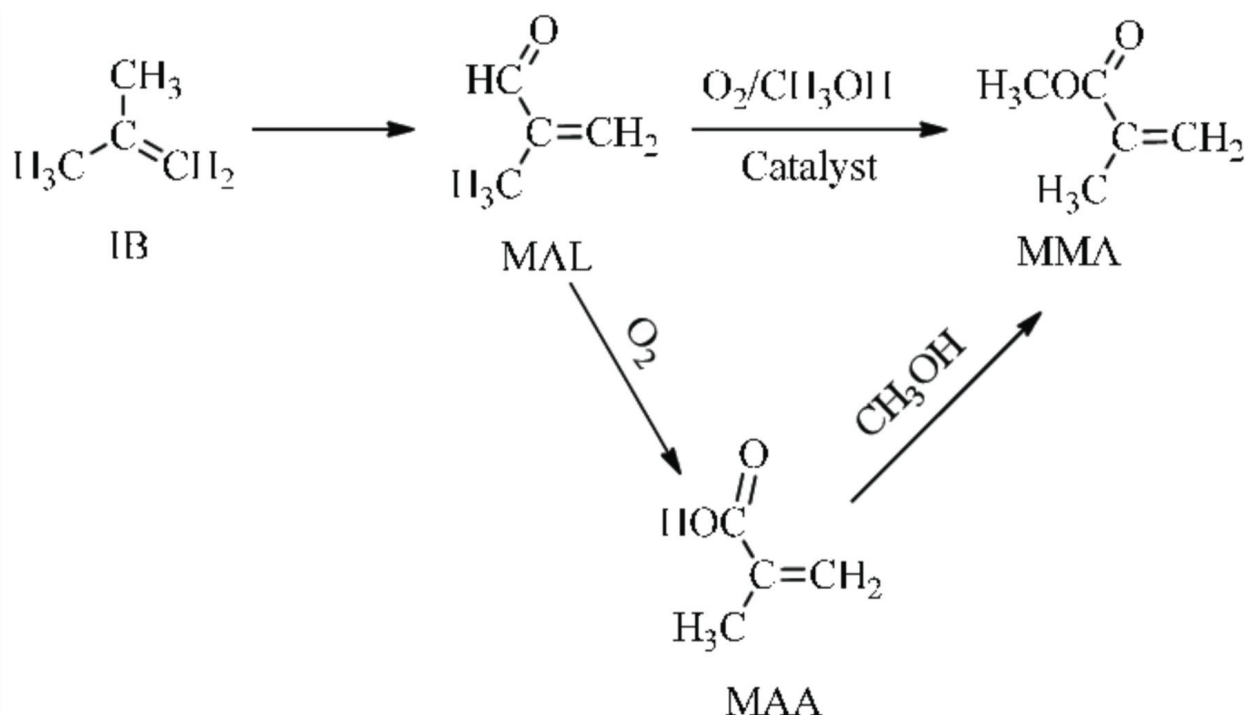


Figure 3: Partial oxidation of a C4 stream to give Methacrylic Acid

2.3.2 ACH (Acetone Cyanohydrin) Process

The ACH process converts acetone to acetone cyanohydrin (ACH) by reaction with hydrogen cyanide (HCN). ACH is then converted (hydrolysis followed by further steps) to methacrylamide or methacrylic derivatives and ultimately to MAA or MMA. Historically, the ACH route has been a major commercial route for methyl methacrylate (MMA) and related methacrylate chemicals. (Curtis, I. Carlson et. al., 2003)

Typical catalysts & conditions

1. ACH formation: acetone + HCN in the presence of a basic catalyst (phase transfer or base). Conditions are typically ambient to mild exothermic conditions; HCN handling and scrubbing systems are essential.
2. ACH hydrolysis / conversion: strongly acidic hydrolysis (traditionally sulfuric acid), or modern heterogeneous catalysts to avoid large quantities of waste acid. Temperatures and pressures vary with specific plant designs.

Yields & issues

The ACH route can give high yields to downstream methacrylic products when optimised. However, the process involves hydrogen cyanide—a highly toxic and hazardous reagent—and forms by-products such as acetone sulfonic acids, formamide (from HCN hydrolysis), and requires careful waste management.

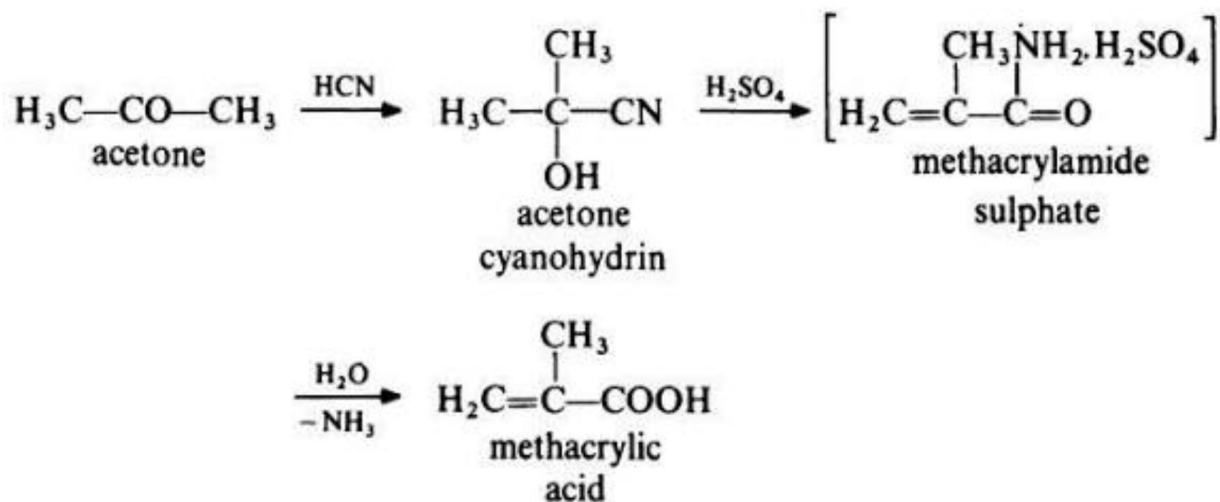


Figure 4: ACh process for Methacrylic acid production

2.3.3 Oxidative Dehydrogenation / Dehydration of Isobutyric Acid (IBA)

Isobutyric acid (IBA) can be oxidatively dehydrogenated/dehydrated to methacrylic acid in the gas phase over selective oxidation catalysts.

Typical catalysts & conditions

1. Catalysts: molybdenum- and vanadium-containing mixed oxides, heteropoly acids supported on silica, and modified molybdophosphoric materials.
2. Temperatures: typically 300–450 °C for gas-phase oxidative dehydrogenation.
3. Pressure: atmospheric to a few bar.

Yields & selectivity

Reported selectivities for IBA → MAA are moderate—literature and patents report selectivities in the

range of roughly 60–75% under certain conditions. Side reactions and over-oxidation reduce overall selectivity and must be managed by catalyst choice and reaction conditions.

2.4 Comparison of the Three Processes

Table 6: Comparison of Syntheses routes for Methacrylic acid

Process Variable	C4 Oxidation	ACH Process (Acetone Cyanohydrin)	IBA Dehydration / Oxidative Dehydrogenation
Pressure	Near atmospheric – 3 bar	1–2 bar (ACH formation), hydrolysis may use slightly higher pressure	Atmospheric – 2 bar
Temperature	200–450 °C (stepwise for oxidation)	0–50 °C for ACH formation; 60–120 °C for hydrolysis	300–450 °C
Catalyst	Vanadium-phosphorus oxides, Mo–V–P mixed oxides, supported V oxides	Acid catalysts (sulfuric acid, strong solid acids)	Molybdenum-vanadium mixed oxides, heteropoly acids
Reagent	Crude C4 stream (isobutene/isobutane), air/oxygen	Acetone + HCN (hazardous)	Isobutyric acid, O ₂ / air for oxidative route
Reaction time	Seconds to minutes	Minutes to hours	Minutes (gas-phase)

	(continuous gas-phase)	depending on step	continuous)
Percentage yield	70–90% (to MAA, depending on selectivity)	80–95% (stoichiometric, high yield)	60–75% (moderate selectivity)
Impurities / By-products	CO, CO ₂ , maleic anhydride, polymeric tars, methacrolein residues	Acetone, ammonium salts, residual HCN, acetone sulfonic acid	CO _x , overoxidation products, polymeric residues

2.4.1 Analysis: C4 Oxidation Route

Advantages:

1. Compatible with crude C4 streams available at steam crackers (butenes, butadiene-rich cuts).
2. Can be configured as a continuous gas-phase oxidation process with high throughput.
3. Mature catalyst families (V–P, V–Mo, Mo–V–P) exist and have been optimised for selective partial oxidation steps.

Disadvantages and possible side reactions:

1. Over-oxidation to CO and CO₂ (complete combustion) reduces selectivity.
2. Formation of by-products such as maleic anhydride, crotonaldehyde, acetaldehyde, and polymeric tars depending on feed composition and conditions.
3. Partial oxidation may produce methacrolein (MAL) as an intermediate—controlling the two-step oxidation is critical to avoid accumulation and side reactions.
4. Catalyst deactivation (coking, phase changes) and need for regeneration.

5. Presence of butadiene and other dienes in crude C4 can lead to undesired reactions (polymerisation, condensation) and require upstream purification.

2.4.2 Analysis: ACH Process

Advantages:

1. Historically proven route to MMA/MAA with high potential yields when optimised.

Disadvantages:

1. Requires hydrogen cyanide (HCN): highly toxic, poses acute safety hazards and complex environmental controls.
2. Side-products and acid wastes (if sulfuric acid used) can generate disposal and corrosion issues.
3. Modern improvements (heterogeneous catalysts for hydrolysis) reduce some waste but do not eliminate HCN hazards.

2.4.3 Analysis: Isobutyric Acid (IBA) Oxidation

Advantages:

1. Direct conversion from a C4 carboxylic acid; suitable where IBA feedstock is available.

Disadvantages:

1. Requires high temperatures (300–450 °C) and selective catalysts.
2. Selectivity to MAA is moderate (often 60–75%) with competing over-oxidation and by-product formation.

2.5 Feedstock Availability for Methacrylic Acid in India

Dahej is a major petrochemical hub with steam cracker and petrochemical complexes (OPaL/ONGC Petro additions and other operators) producing ethylene, propylene and C4 fractions as coproducts.

Crude C4 streams (containing isobutene, butenes, butadiene, and butanes) are commonly produced at crackers and can be sourced from cracker operators, refineries, and C4 traders in the Dahej PCPIR/Dahej industrial cluster. Proximity to these crackers makes Dahej a favourable site for C4 oxidation processes.

2.6 Final Selection and Justification

Based on catalyst options, yields, continuous-operation compatibility, and feedstock availability in petrochemical clusters such as Dahej, the **C4 oxidation route** is selected for bulk production of methacrylic acid.

1. The C4 oxidation route integrates well with steam cracker crude C4 availability at Dahej, minimising feedstock logistics and cost.
2. It can be operated continuously with well-studied oxidation catalysts to achieve economically favourable yields and throughput.
3. The ACH route was rejected primarily because it requires hydrogen cyanide (HCN), which is highly hazardous and requires stringent safety measures and complex waste handling.
4. IBA oxidation was not selected due to moderately lower selectivity and the requirement for high-temperature operation and specialised catalyst systems.
5. The global competitors for methacrylic acid are also shifting to the C4 oxidation route due to the advantages of this process. Hence if we want to establish a plant for the long run, it is better to go for the C4 route.

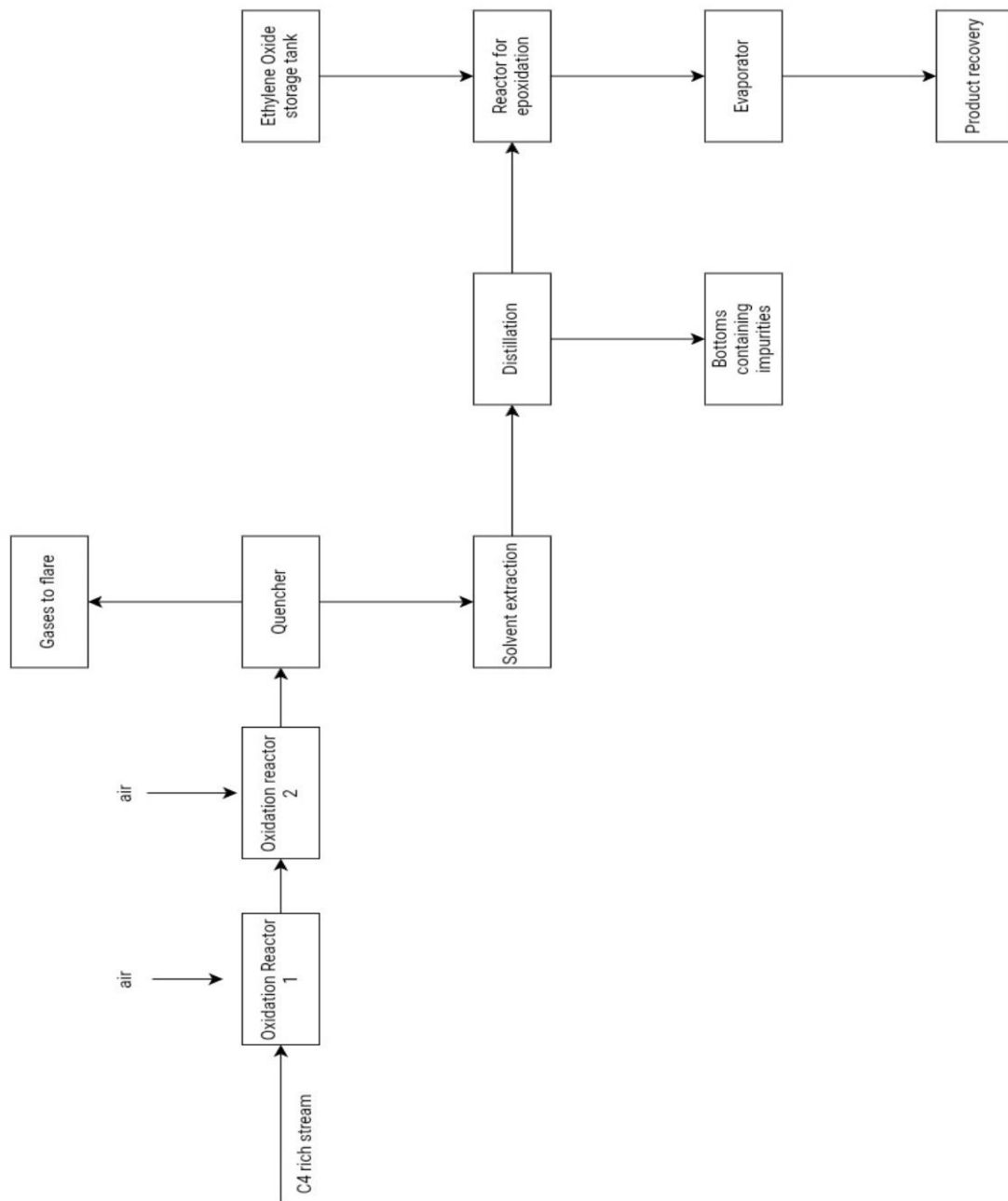


Figure 5: Block Flow Diagram for syntheses of HEMA and MAA

Chapter 3: Site Selection

3.1: Factors for Site Selection

For a methacrylic acid (MAA) and 2-hydroxyethyl methacrylate (HEMA) plant, site selection is a critical strategic decision due to the hazardous, bulk nature of raw materials and the multi-step chemical processes involved. The core criteria are:

- **Proximity to Petrochemical Feedstock:** The catalytic oxidation process for MAA requires a reliable supply of C4 hydrocarbons and oxygen. Siting the plant near large petrochemical complexes minimizes feedstock transport costs and ensures operational continuity. For HEMA, access to methacrylic acid is essential for the epoxidation step; co-location within an established methacrylate cluster enables efficient intermediate handling.
- **Logistics Infrastructure:** MAA and HEMA plants demand robust inbound and outbound transport networks (road, rail, port) to handle bulk chemical supply and enable timely export of finished products. Integrated clusters with efficient connectivity help minimize delays and logistics expenses.
- **Utilities and Steam Availability:** These processes are energy-intensive, needing reliable electricity and process steam. Clusters like Dahej and Vapi have existing utility networks, and proximity to large chemical sites allows for possible steam purchase rather than captive generation.
- **Water Supply and Effluent Management:** Large quantities of water are required for reaction, cooling, and cleaning. Sites must have a sustainable water source and advanced effluent treatment facilities due to the hazardous nature of organic acids and solvents.
- **Land and Regulatory Compliance:** Sufficient land for plant, expansion, safety buffer zones, and environmental approvals. Gujarat and Maharashtra offer streamlined chemical sector land acquisition and clearances.
- **Skilled Labor and Industrial Ecosystem:** Availability of trained chemical engineers and technicians, plus proximity to suppliers of catalysts, solvents, and equipment.
- **Ease of Doing Business:** Favorable regulations, industrial cluster support services, and ease of obtaining hazardous chemical permits all impact setup and ongoing operations.
- **Cluster Synergy:** Co-location with allied industries (such as resins, adhesives, paints) in clusters facilitates feedstock availability, shared services, and downstream integration.

Justification for Exclusion of Certain Parameters

- Proximity to end-user market is less critical: MAA/HEMA are shelf-stable, and Indian demand is spread nationally; clusters are connected to export ports.
- Climate is not a major factor as chemicals degrade minimally under standard storage, and clusters have adequate warehousing.
- Effluent processing is ensured by cluster-scale treatment facilities; individual plants do not need to independently construct ETPs unless mandated.
- Housing and labor supply are sufficient in semi-urban chemical hubs, and relocation/hiring logistics are straightforward.

3.2: Site Comparison and Selection with Justification

Table 7: Site comparison

Parameter	Dahej PCPIR, Gujarat	Ankleshwar, Gujarat	Vapi, Gujarat	Thane, Maharashtra
Petrochemical Complex	GNFC, Reliance, ONGC	GNFC, Narmada Chemicals	Diverse Chemicals, SME units	Deepak Fertilizers, legacy clusters
Transport Links	Major port, rail, road	Rail, road, moderate port	Rail, road, Mumbai port	Rail, road, Mumbai port
Utility Cost	Low, cluster tariffs	Low, cluster tariffs	Moderate, SME rates	Higher tariffs
Steam Availability	Cluster steam supply	Cluster steam, captive	Cluster steam, captive	Captive only
Water Cost (Rs/kL)	15–18	18–20	19–22	24–28
Land Cost (Lakh/acre)	16–22	13–20	15–24	38–45
Labor Cost	480	490	520	560

(Rs/day)				
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Table 8: Weightage of each parameter

Parameter	Weight (0-10)
Petrochemical complex	10
Transport links	9
Utility/steam	8
Water cost	7
Land cost	8
Labor cost	5
Ease of doing business	8
Cluster synergy	8

Table 9: The grading of Sites Based on Parameters

Parameter	Dahej	Ankleshwar	Vapi	Thane
Petrochemical complex	10	9	8	7
Transport links	10	8	8	8
Utility/steam	9	8	8	6
Water cost	9	8	8	6
Land cost	8	9	7	5

Labor cost	7	7	7	8
Ease of doing business	9	9	7	7
Cluster synergy	10	9	9	8

Table 10: Weighted Average of Parameters

Weighted Parameter	Dahej	Ankleshwar	Vapi	Thane
Petrochemical complex	100	90	80	70
Transport links	90	72	72	72
Utility/steam	72	64	64	48
Water cost	63	56	56	42
Land cost	64	72	56	40
Labor cost	35	35	35	40
Ease of doing business	72	72	56	56
Cluster synergy	80	72	72	64
Total weighted score (out of 10)	6.9	6.6	5.9	5.0

Based on the weighted score comparison, **Dahej PCPIR** in Gujarat emerges as the optimal location for setting up a methacrylic acid and 2-hydroxyethyl methacrylate plant. This site offers direct access to

petrochemical feedstocks, export ports, cluster steam/utilities, cost-effective land, and a mature chemical manufacturing ecosystem, making it ideal for specialty monomer production and expansion.

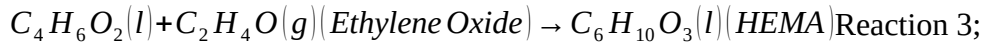
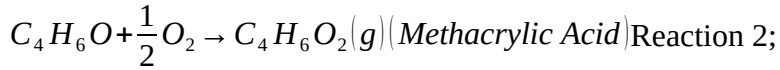
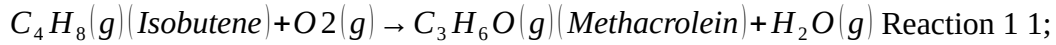
Chapter 4: Thermodynamic Feasibility

The thermodynamic feasibility of any process is characterized by the Gibbs free energy of that process at the given condition. The Gibbs free energy or ΔG for any process should be negative for the process to be feasible. The standard Gibbs free energy or $\Delta^\circ G$ is the ΔG at 25 °C or 298 K. The Table below lists the standard Gibbs energy of formation of the main species involved in the synthesis of HEMA using the chosen process. (All the thermodynamic values have been taken from Yaws C.L. , "Physical Properties Handbook", McGraw Hill publications)

Table 11: Standard Gibbs energy of formation

Species	$\Delta^\circ G$ formation (kJ/mol)
Isobutylene	58.07
Methacrolein	-57.6
Methacrylic Acid	-287.72 for gas phase, -203.65 for liquid phase
Water	-13.1
Ethylene Oxide	-291.81
2-Hydroxyethyl Methacrylate	-228.64

The reactions occurring for forming HEMA are as follows-



The ΔG° of any reaction is given by the formula

$$\Delta G^0 = \sum v_i G_{fi}$$

where v is the stoichiometric coefficient of species “i” and G_f is the Gibbs energy of formation for the species.

Using this relation, we obtain the Gibbs energy of formation at standard conditions for all the three major reactions. The results are in the table given below-

Table 12: Gibbs energy of formation for the reactions

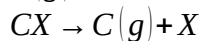
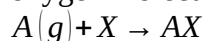
Reaction no.	Gibbs energy (kJ/mol)
1	-344.31
2	-230.12
3	-75.06

Since for all three reactions the ΔG° is negative, we conclude that all the reactions are thermodynamically feasible. Since the ΔG° is not zero, the reactions are not limited by equilibrium.

Chapter 5: Kinetics of the Reaction

5.1 Partial oxidation of Isobutene and Methacrolein

The partial oxidation of both Isobutene and Methacrolein follow the Mars-van Krevelen mechanism. The Mars-van Krevelen mechanism is when the hydrocarbon gas molecule reacts with the adsorbed oxygen molecule in the reactant pore. The mechanism is as follows-



where, A represents Oxygen, X represents the vacant site on the catalyst, B represents the hydrocarbon species and C is the product molecule. The reaction has a characteristic factor of ϕ which the desorption rate of Oxygen from the catalyst site.

5.1.1 Kinetic parameters for partial oxidation of Isobutene to Methacrolein

Molybdenum based catalysts provide the highest selectivity for this reaction. Based on the work done by Hoornaerts et. Al in their 1997 paper titled “Modification of kinetic parameters by action of oxygen spillover in selective oxidation of isobutene to methacrolein” the rate equation for this reaction is given as

$$r_{MAC} = \left(\frac{k_{red} P_{C4}^x k_{ox} P_{O2}^y}{k_{red} P_{C4}^x + k_{ox} P_{O2}^y} \right)$$

The values of the kinetic parameters are given in Table 13.

Table 13: Kinetic parameters for reaction 1, Benyahia et. al

Temperature (K)	k_red	k_ox	X	Y	ψ
	(mol atm ⁻¹ s ⁻¹ m ⁻²)	(mol atm ⁻¹ s ⁻¹ m ⁻²)			
623	8.88e-11	0.92e-11	0.03	0.7	0.16

643	16.82e-11	1.72e-11	0.25	0.77	0.09
663	26.6e-11	3.47e-11	0.33	0.7	0.12

The activation energy for catalyst reduction is 52 kJ/mol and for reoxidation is 142 kJ/mol. The constant K0 for the kinetic parameter is chosen to the parameters at 643 K due to the low psi value and Arhenius equation is used to calculate the values at any temperature.

5.1.2 Kinetics parameters for the partial oxidation of Methacrolein

Heteropoly acid catalysts are used for facilitating these reactions. They are usually phospho-Molybedate complexes. The rate equation for the oxidation of Methacrolein is given by

$$-r = \frac{K_{O_2} P_{O_2}^n (1 + P_{H_2O}^m)^2}{K_{O_2} P_{O_2}^n (1 + P_{H_2O}^m) + 0.5 K P_{MAC}}$$

where, n = 0.7, m = 0.65 and the values of the kinetic parameters are described in the following table

Table 14: Kinetic parameters for reaction 2

Kinetic Parameter	Value (mol/hkgcatPa ^x)	Activation energy (kJ/mol)
Ko2	5.35E-06	99.99
K	7.6E-04	154.4

5.2 Esterification of Methacrylic acid with Ethylene Oxide

The rate equation for this reaction is given by

$$-r_{EO} = \frac{K_1 K_2 k_3 [EO][MA]}{1 + K_1 [EO] + K_1 K_2 [EO][MA] + K_4^{-1} [HEMA]}$$

where, [Cx] is the concentration of the species “x” and the value of the kinetic parameters are given in the table below.

Table 15: Kinetic parameters for reaction 3

Kinetic Parameter	Value at 50 °C	Value at 60 °C	Value at 70 °C	Value at 80 °C	Value at 90 °C	Activation energy (kJ/mol)
K1	6.654	3.909	2.368	1.478	0.944	33.53

K2	0.396	0.368	0.344	0.322	0.304	47.62
k3	0.0887	0.183	0.459	1.095	2.488	54.11
K4	0.08	0.19	0.428	0.921	1.901	77.21

The catalysts for this reaction include iron or chromium based catalysts. The use of a tertiary amine also helps in promoting this reaction.

5.3 Oxidation of Isobutene, Methacrolein and Methacrylic acid to side products

Chapter 6: Material Balance

Material balance is essentially applying the law of conservation of mass to the reactor system. We can do an overall balance for the reactor by using the known conversion values. The conversion of Isobutene in reactor 1 is 99.6% and the yield of Methacrolein is 79.6%. Similarly for reactor 2, the conversion of methacrolein is 78% and the selectivity is 85%. Low conversions of methacrolein are preferred since at conversions higher than 78%, the selectivity to Methacrylic acid decreases. For the esterification, the conversion with respect to methacrylic acid is 99.5%. Here, ethylene oxide is the limiting reactant due to its high reactivity.

For given production rate of 8,000 tonnes per annum, we have the production rate of HEMA to be 1 tonnes per hour or 19.23 kmol/h. Using this we back calculate the amount of Isobutene